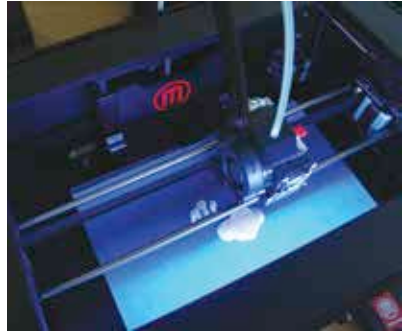


Step 5: The next step is very easy – the machine proceeds to process the STL file and fabricate the object that's been designed. For most consumer grade 3D printing machines and most designs, the entire printing process is automated. Only in certain rare cases, manual intervention may be called for; E.g. If the printing process requires large material quantities and reloading is necessary or if parts of the design need. The printer creates layers measuring 0.1mm in average thickness. Based on the material, this can be thicker or thinner. Printing objects can take a variable amount of time – from minutes to hours to even days. You may sometimes be required to keep checking in on the printer's progress to ensure that there are no errors or misalignments.



The printer apparatus moves in all dimensions to fabricate the CAD-based object

Step 6: Once the object has been printed, its removal from the printer is an extremely delicate and critical step. In many cases, the printing process leaves the object's surface hot and malleable, and in certain cases requires additional time to clear off fumes and particulates. Users are advised to take special precautions such as wearing gloves and glasses when removing the object from the printer.

Step 7: The next stage involves processing the item. With most 3D printers, the final object is usually found covered with the remains of the additive materials, or a layer of powder or coarse material. The processing stage requires either dusting off the physical particulates or bathing the object in water to remove water soluble elements. It's important to note that not all objects taken from a 3D printer are immediately ready for processing. Depending on the material and design configuration used, some time may be required for the additive material to "cure" and harden completely before it can be processed without risk to the physical integrity of the object. If this is ignored, there's a significant risk that certain parts of the object will fall apart, dissolve or weaken the overall structure of the object.

The above stated workflow of 3D Printing is common to all models. Once the process is complete, you can use the final object for its intended purpose.